

Cybersecurity Concepts Lab for Teachers

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FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-inked, in the order the lessons take them.

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OCTOBER 2026

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Demo 10: Backups and Ransomware Recovery (The Lose-Your-Lunch-Box Game)

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE

Big idea	Things go wrong: spills, losses, mistakes and attacks. A good backup is a second (and third) copy, kept somewhere else, that you have tested . Backups are the strongest defense against data loss, including ransomware.
Grade bands	K-2 (the toy box story), 3-5 (full Lunch Box game), 6-8 (full + 3-2-1 rule + sync vs backup), 9-12 (full + ransomware lifecycle, immutability, recovery planning)
CSTA-style concepts	Computing Systems (storage, troubleshooting), Networks and the Internet (cybersecurity: data protection) (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Envelopes, index cards, a cheap padlock or paper clips as "locks", dice. No devices.
Teachers do it first	Yes. Part A about 20 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Silently think of one digital thing you would hate to lose. Where is its second copy? Is that copy in the same place or on the same account? When did you last restore it? (Do not share your answers; note one improvement.)

01 The big idea

EVERYTHING YOU CARE ABOUT in digital form can disappear: a phone dropped in water, a laptop stolen, a mistaken delete, a failing drive, a fire, or **ransomware** (malicious software that locks your files and demands payment to unlock them). The antidote is the same for all of them: **another copy, somewhere else, that you know works.**

The lunch-box game makes it concrete. Your lunch box holds precious stuff (homework, photos, game saves). Events strike. Students choose backup strategies *before* they know which events will happen, just like real life.

The lesson's three sentences:

1. **If it only exists in one place, it is not safe.**
2. **A backup you have never restored is a hope, not a plan.**
3. **A copy that gets damaged together with the original is not a backup.**

02 What a teacher should know (plain language)

2.1 WHAT IS A BACKUP?

A **backup** is a separate copy of data, kept so you can get your data back after loss, damage or attack. Key properties:

- **Separate.** Not on the same device or in the same place as the original, or it fails with it.
- **Recent.** The more recent, the less you lose.
- **Tested.** You have practiced restoring.
- **Protected.** Backups can be stolen too; use account security (Demo 6), encryption, and access control.
- **Versioned.** Keeping older versions lets you go back before a mistake or damage was copied.

2.2 BACKUP VERSUS SYNC (A VERY COMMON CONFUSION)

Sync (cloud storage that mirrors your folder across devices) keeps copies *the same*. If you delete a file, or something corrupts or locks it, the sync often copies that change everywhere. Many sync services keep a recycle bin or version history for a time, so they can **help**, but check their rules [VERIFY by provider]. A **true backup** keeps a separate history that doesn't get silently overwritten. Rule of thumb: *Sync is for convenience; backup is for rescue.*

2.3 THE 3-2-1 RULE (AND ITS UPGRADED FORM)

- **3** copies of the data (the original plus two backups).
- **2** different types of storage (for example, the laptop plus an external drive, or a drive plus a cloud service).
- **1** copy **off-site** (away from the building: cloud, or a drive at a relative's home).

Many security teams now add: **1** copy **offline or immutable** (disconnected, or set so it cannot be altered or deleted for a time) and **0** errors after **verifying** by restore tests (sometimes written 3-2-1-1-0) [attribution not verified; the original 3-2-1 rule is often credited to photographer Peter Krogh]. Teach the simple form to students and the upgraded form to 9-12.

2.4 RANSOMWARE IN PLAIN LANGUAGE

Ransomware is malicious software that makes your files unreadable (by locking them with encryption, see Demo 4) and shows a message demanding payment, usually in cryptocurrency. Many modern groups also **steal a copy** of the data first and threaten to publish it ("double extortion"), so a backup solves the loss problem but not the leak problem [general description; CISA's #StopRansomware Guide says to keep offline, encrypted backups and test them because many ransomware variants try to find and delete or encrypt accessible backups, VERIFIED 2026-10-05: cisa.gov/stopransomware/ransomware-guide].

Teachers should know:

- **How it usually arrives** (conceptually): phishing messages with bad links or attachments (Demo 3), unpatched software (Demo 7), weak or reused remote-access passwords (Demo 1), and fake downloads (Demo 11). We do not describe how to build or deploy it.
- **Backups are the main way to recover without paying**, if they were kept offline or protected from the same attack. Attackers often try to find and destroy backups.
- **Should you pay?** Law enforcement agencies advise against paying: there is no guarantee you get your data back, and payments fund further crime [the exact FBI and CISA wording could not be read by the checker in Round 3B (fbi.gov blocked; the CISA ransomware guide page read does not state it), so keep this as general guidance and cite the agency page when you present it]. Some victims do pay after weighing options with experts, which is why the decision belongs with professionals, not with a child or a teacher alone.
- **If it happens:** disconnect the device from the network (unplug the network cable or turn off Wi-Fi), do not pay or contact the criminals, tell a trusted adult (at school: tell IT immediately), and report it (in the US:

the FBI IC3 and CISA; see StopRansomware.gov; the CISA ransomware guide page was read on 2026-10-05). Organizations have an incident response plan (Demo 13).

- **Schools are targets.** Ransomware has repeatedly hit US K-12 districts, causing closures and data leaks [widely reported; the CISA/FBI/MS-ISAC K-12 reports were not read in Round 3B]. This is a reason for calm preparedness, not fear.

2.5 BACKUPS IN THE REAL WORLD (SHORT GUIDE)

- **Phones and tablets:** turn on the built-in cloud backup and protect that account with a strong password and a second lock (Demo 6).
- **Computers:** built-in tools (File History, Time Machine) to an external drive, plus a cloud backup service; keep the drive disconnected when not backing up.
- **Photos, school work and family documents:** at least two homes, one off-site; keep a printed list of accounts and recovery methods.
- **Test:** once or twice a year, restore one file and open it.

2.6 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"My cloud storage is my backup."	Sync is not backup. Check version history and deletion rules.
"I have a backup, so I'm safe."	Only if it's recent, protected and tested.
"A copy on the same computer counts."	If the computer fails or is locked, so does the copy.
"Ransomware only hits big companies."	Individuals, schools and small businesses are hit too.
"Paying fixes it."	Not guaranteed; encourages more attacks.

03 Materials (zero to low cost)

- A lunch-box envelope per team with six **food cards** (Homework, Photos, Art project, Music, Game save, Family recipes); copy cards (students draw or photocopy their own).
- **Backup place cards:** Same Bag; Other Room (connected: external drive or synced folder); Far Away (cloud with version history, or another home); Locked Cabinet (offline copy).
- **Event cards** (Section 4.2), padlocks or paper clips marked "LOCKED" (paper only, no real encryption), timer, score sheet, sticky-dot tokens.

04 The Lunch Box Game

4.1 SETUP

EACH TEAM (3-4 STUDENTS) is a "family" with a **lunch box** holding six precious "food cards". Teams have **6 tokens** to spend on a backup plan. They choose where to store copies:

BACKUP OPTION	COST	WHAT IT PROTECTS AGAINST
Same Bag (copy in the same bag)	1 token	Mistaken delete only
Other Room (a copy that stays connected: a drive left plugged in, or a synced folder)	2 tokens	Spills, lost bag, device failure (but not ransomware)
Far Away (another school, relative's house, cloud)	3 tokens	Fire, theft, loss of the whole building
Locked Cabinet (an offline copy in the building, not connected)	2 tokens	Ransomware and sync problems (but not fire)

The listed cost buys **one copy of one food card** (so 6 tokens is a limited plan: for example, one Far Away copy plus one Locked Cabinet copy, or six Same Bag copies). Teams decide which cards to protect most. In the game, Far Away means a backup service that keeps version history. **Teams must write their plan on a slip before any event cards are drawn.**

4.2 EVENT CARDS (DRAW ONE PER ROUND; 6 ROUNDS)

#	EVENT	EFFECT
1	Spilled juice	Everything in the box (original and any copy in the same bag) is ruined.
2	Lost bag	The box and anything in the Same Bag are lost. Other Room and beyond survive.
3	Oops, deleted!	One chosen food card is erased. Any copy anywhere restores it.
4	The Lunch Locker (ransomware)	The box, Same Bag and Other Room copies (all connected) are padlocked. Only Far Away (versioned) and Locked Cabinet copies survive.
5	Cafeteria fire	Everything in the building is lost (the box, Same Bag, Other Room, Locked Cabinet in the building). Only Far Away copies survive.
6	Sneaky swap: a food card was silently replaced with a wrong card last week.	Only teams with an older version in Far Away or Locked Cabinet can get the right one.
7	Power cut	No effect on cards, but teams need access to the Other Room copy in 60 seconds (timed).
8	Friend borrowed and broke it	The Other Room copy is damaged.
9	Surprise test day	The teacher runs a Restore Drill (see below).
10	Quiet day	Nothing happens. Reflect on luck: "Was it a good plan or good luck?"

TEACHER TIP

*before the game, privately mark which team's backup of one card is **blank** (a backup that failed). In the Restore Drill, that team finds out why "tested" matters.*

4.3 THE RESTORE DRILL (THE MOMENT OF TRUTH)

Draw the Surprise test day card, or run it after the six rounds. For each team, the teacher says: "Restore your Game save and Photos in 90 seconds." Teams retrieve their copies. If a copy is blank, mismatched or missing (a "bad backup"), they lose that card. Debrief: "Who tested before the drill? Who found a surprise?"

4.4 SCORING

- 1 point per food card **still readable and correct** at the end.
- +2 points for a team that restored within the time limit.
- +1 point for a team that can **explain** which events their plan protected against.
- Do not reward luck: if two teams scored the same but one had no plan, discuss that in debrief.

05 Part A: Teachers do it first (about 20 minutes)

1. **Plan (4 min).** Spend your 6 tokens and write a plan slip.
2. **Play (8 min).** Draw three events and check what survives.
3. **Restore Drill (3 min).** A colleague secretly blanks one of your backup cards; restore in 90 seconds.
4. **Real life (5 min).** What is your most irreplaceable digital item? Where are its copies and when did you last restore it? Note one improvement; do not share details.

06 Part B: Teaching it

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook: "Lose your lunch" story; show a lunch box with six cards.
2-5	Teams spend 4 tokens (shorter version) and write plan slips.
5-11	Draw four events (spill, lost bag, locker, fire). Teams remove lost cards and score.
11-14	Debrief: which place did most work? Sync vs backup.
14-15	Exit slip: "A backup is good if it is _____, _____ and _____."

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-6	Hook, vocabulary, 3-2-1 rule.
6-12	Teams plan.
12-30	Six events with the Restore Drill. Teacher uses the "blank backup".
30-37	Debrief: sync versus backup; ransomware overview; the "don't pay, tell, disconnect" steps.
37-43	Redesign: teams make a new plan with 8 tokens for a family with a new need (e.g., a small business) and share.
43-45	Exit ticket and handout.

07 Teacher script

Hook (2 minutes). "I've got a lunch box. Inside, cards for your homework, your photos, your game save, your art, your music, your family recipes. Everything you love is in there. Now: what could happen to a lunch box today? Spill, lost, dropped, stolen. Today we find out what a smart plan looks like."

During events. "Spilled juice! Take everything out of the box and the bag. What survives? Lost bag! What survives? Notice a copy in the same bag didn't help."

The Locker (ransomware) card. "A scary screen says: 'Your lunch is locked! Pay us!' What do you do? Grown-up first, don't pay, don't answer. If you have a locked cabinet copy, you can restore. That's why we keep one offline. Real ransomware is more complicated, but the lesson is the same: backups make you hard to blackmail."

Close. "A backup is a second copy, somewhere else, that you know how to get back. Today your family can pick one important thing and make a second copy, then practice. If something ever gets locked or lost, tell a trusted adult. It's not your fault; it can happen to anyone."

08 Kid-friendly analogies by band

BAND	ANALOGY AND ACTIVITY
K-2	The toy box and the drawing. If you lose a favorite drawing, a copy in a different place saves the day. Make a copy of a drawing and put it with a grown-up. "Two homes for special things."
3-5	The Lunch Box game (simplified: 3 tokens and 3 events). Learn "same bag is not safe", "far away is safe from fire".
6-8	3-2-1 and sync versus backup. Full game with cabinet and far-away copies; a ransomware card; restore drill. Discuss "backups should be tested".
9-12	Resilience planning. Recovery time and recovery point objectives (how long to restore, how much work can be lost), immutability, backup security, ransomware lifecycle and decision-making, incident response roles (Demo 13).

09 Discussion questions

1. Which backup place saved the most food cards, and which the least? Why did the Same Bag copy fail on the juice and lost-bag cards?
2. What is the difference between sync and backup? Which helps after an accidental delete, and which after ransomware?
3. Why is it important to test a backup?
4. (9-12) How do you decide what data needs the strongest backup? What is the cost of losing a day of work versus a week?

10 Level-up challenge

- **Family Backup Plan.** Draw a one-page plan: three irreplaceable items, where the original lives, where two copies live, who is responsible, how often it is tested.
- **Recovery time math.** If a restore moves 100 GB at 50 MB per second, how long? [Answer: 2,000 seconds, about 33 minutes.] How long for 2 TB? [About 11 hours.] Why does that matter for planning? (State your rounding and assumptions.)
- **Ransomware ethics debate (9-12).** Should an organization ever pay? Present both views, consult public guidance [VERIFY], and include the effect on victims and criminals.

11 Safety and ethics notes (defense only)

- **Never create, simulate with real software, or demonstrate ransomware or any malware.** The "locker" in this game is a padlock on a paper card. Do not encrypt or lock any real files, including in class demonstrations.
- **Do not encourage students to try "recovery tools" or download software on school or family devices** without an adult's approval.
- **Do not frame ransomware as a "game" students could try.** The learning goal is preparedness.
- **Sensitive backups.** Do not ask students to back up or share personal photos or documents in class.
- **Anxiety.** Some students may have lost precious files or experienced a family loss. Keep tone matter-of-fact: "Anyone can lose things, and planning ahead helps."
- **Disclosures.** If a student says someone threatened them with releasing photos or files, follow your school reporting process (see the Index box); do not ask for the files or screenshots.

12 Assessment idea

- **Exit ticket.** "Name three things that make a backup good, and one reason a copy in the same bag fails." Rubric: 3 = separate, recent/tested/protected and a correct reason; 2 = two correct items; 1 = "copy it".
- **Scenario sorting.** Cards: "File deleted by mistake", "Laptop stolen", "House fire", "Ransomware". Students match each to which copies would help.

13 Sources and [VERIFY] list

- CISA and FBI #StopRansomware guidance; StopRansomware.gov [CISA ransomware guide VERIFIED 2026-10-05; K-12 reports not read].
- CISA and NIST guidance on data backup and recovery (NIST SP 800-184 "Guide for Cybersecurity Event Recovery", December 2016) [VERIFIED 2026-10-05: csrc.nist.gov].
- Peter Krogh, "The DAM Book" for the 3-2-1 rule; US-CERT 2012 backup guidance [not checked].
- FBI IC3 annual report on ransomware complaints [VERIFIED 2026-10-05: the 2025 report counts more than 3,600 ransomware complaints with reported losses above \$32 million].
- Provider documentation on cloud sync version history and recycle bin limits [VERIFY per provider].

Student Handout: The Lunch Box Game

Name: _____

Backup rules

- 3 copies, 2 kinds of places, 1 far away (3-2-1).
- A backup is only good if it is **separate, recent, tested and protected**.
- **Sync is not backup**. Deleting or locking can sync everywhere.

Which places would have saved you?

PLACES: A SAME BAG, B Other Room (stays connected), C Far Away (with version history), D Locked Cabinet (offline, in the building). For each event, write **all** the letters that would have saved your food card.

EVENT	PLACES THAT SAVE YOU
Spilled juice (box and Same Bag ruined)	_____
Lost bag (box and Same Bag gone)	_____
Cafeteria fire (everything in the building lost)	_____
The Lunch Locker (ransomware: all connected copies locked)	_____

Write your reasoning for one of them: _____

My family backup plan

Most important thing: _____

Copy 1 lives: _____ Copy 2 lives: _____ Far away copy: _____

Last time we tested it: _____

If a scary screen says "pay or lose your files"

1. **Pause.** Do not click or call.
2. **Tell** a trusted adult right away.
3. A grown-up can disconnect the device and ask for help.
4. It is **not your fault**.

Remember

A BACKUP YOU HAVE never restored is a hope, not a plan. Practice once!

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 10 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Which places would have saved you (using the game's rules):

EVENT	PLACES THAT SAVE YOU
Spilled juice	B, C, D (anything outside the box and the Same Bag)
Lost bag	B, C, D
Cafeteria fire	C only (the Locked Cabinet is in the building)
The Lunch Locker (ransomware)	C (versioned) and D; not A or B, because connected copies are locked too

Accept any answer that matches the reasoning. The point: no single place covers everything, which is why we want more than one kind of copy.

Recomputed: 100 GB at 50 MB/s = 2,000 seconds (about 33 minutes); 2 TB at 50 MB/s = 40,000 seconds (about 11 hours).

Model limits to say aloud: real ransomware often hunts for and destroys backups it can reach, and many groups also steal data first ("double extortion"), so a backup solves loss but not leaks. Cloud sync alone is not a backup unless it keeps version history you have tested [VERIFY by provider].